NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 10: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Of the two strands of Artificial Intelligence activity, the one that seeks to understand how intelligent behaviour arises is the _____.

A. cognitive approach.
B. engineering approach.
C. machine learning approach.
D. symbolic approach.

Correct Answer: (A)

Q2. A learning agent receives percepts, selects external actions, and uses feedback to improve its behaviour. Which component is primarily responsible for making improvements**?**

A. Critic
B. Learning element
C. Performance element
D. Problem generator

Correct Answer: (B)

Q3. A classifier is trained on one set of labelled examples and subsequently evaluated using previously unseen examples. The main purpose of using the unseen examples is to _____.

A. construct the training model.
B. assess the model's accuracy on new data.
C. increase the number of training attributes.
D. guarantee that the training accuracy is 100%.

Correct Answer: (B)

Q4. Which statement best distinguishes the traditional emphasis of statistics from that of machine learning?

I. Statistics often emphasizes inference, uncertainty quantification and parameter interpretation.
II. Machine learning often emphasizes predictive performance and generalization.

Choose the correct option.

A. Statements I and II
B. Only Statement II
C. Only Statement I
D. Neither I nor II

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Correct Answer: (A)

Q5. In pure inductive inference, given a collection of examples of f, return a function h that approximates f. The function h is called a _____.

A. utility function.
B. hypothesis
C. approximation function
D. classifier

Correct Answer: (B)

For Q6 – Q10

Consider the following Decision Tree learning problem. The objective is to predict the Boolean class **Play?** from the attributes **Outlook** and **Wind**.

ID	Outlook	Wind	Play?
1	Sunny	Weak	No
2	Sunny	Strong	No
3	Overcast	Weak	Yes
4	Overcast	Strong	Yes
5	Rain	Weak	Yes
6	Rain	Weak	Yes
7	Rain	Strong	No
8	Rain	Strong	No

Q6. What is the entropy of the complete training set?

A. 0.000
B. 0.500
C. 0.811
D. 1.000

Correct Answer: (D)

Detailed Solution:

$$Total\ instances\ (N) = 8; Class\ 'yes'(p) = 4; Class\ 'No'(n) = 4$$

$$Probability\ of\ each\ class: P(Yes) = \frac{4}{8} = 0.5; P(No) = \frac{4}{8} = 0.5$$

$$Entropy(H) = -0.5\log_2(0.5) - 0.5\log_2(0.5) = 0.5 + 0.5 = \mathbf{1.000}$$

Q7. If **Outlook** is used as the splitting attribute, what is the weighted entropy after the split**?**

A. 0.250
B. 0.500

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C. 0.750
D. 1.000

Correct Answer: (B)

Detailed Solution:

Sunny branch: 2 total instances and both are "No".

$$H_{Sunny} = 0.0$$

Overcast branch: 2 total instances and both are "Yes".

$$H_{Overcast} = 0.0$$

Rain branch: 4 total instances (2 are "Yes" and 2 are "No").

$$H_{Rain} = -0.5\log_2(0.5) - 0.5\log_2(0.5) = 1.0$$

$$Weighted\ entropy = \left(\frac{2}{8} * 0.0\right) + \left(\frac{2}{8} * 0.0\right) + \left(\frac{4}{8} * 1.0\right) = \mathbf{0.500}$$

Q8. What is the information gain obtained by splitting the training set on **Outlook**?

A. 0.250
B. 0.500
C. 0.750
D. 1.000

Correct Answer: (B)

Detailed Solution:

$$Information\ Gain = Total\ entropy - weighted\ entropy = 1.0 - 0.5 = \mathbf{0.500}$$

Q9. Suppose **Outlook** is selected as the root attribute. Which of the following branches produces a pure subset?

A. Only Sunny
B. Only Rain
C. Sunny and Overcast
D. None of the branches

Correct Answer: (C)

Q10. Suppose the decision tree first tests **Outlook**. For the Rain branch, it subsequently tests **Wind**. What class should the tree predict for a new instance with: **Outlook** = Rain and **Wind** = Weak?

A. Yes
B. No
C. Cannot be determined; Outlook is not Boolean
D. Cannot be determined; Training set is balanced

Correct Answer: (A)

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